SKYG

ARDUPILOT CONFIG

SKYG SPEED 90B

SpeedyBee F405 WING MINI (ArduPilot target: SpeedyBeeF405WING) · ArduPlane 4.5 or later

REVISION C6B-9 · 2026-10-11

OVERVIEW

Starting ArduPlane configuration for SkyG Speed 90B on the SpeedyBee F405 WING MINI (firmware target SpeedyBeeF405WING, ArduPlane 4.5 or later). Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), then work through the bench checks.

Flight controller: on the wing-bay floor, arrow forward; servo BEC jumper set to 6 V for the ES9052MD servos.

WARNINGS

- These are starting values, not a tested setup. Verify every output, direction and endpoint on the bench before the first flight.
- Never raise THR_MAX: it protects the prop's rpm rating.
- No airspeed sensor: the throttle limits do not stop a dive from building speed. Fly stage runs level.
- SkyG Speed 90 was designed around an rpm governor at 29,500 rpm. ArduPilot cannot run the governor script on an F405 board. Until a governor is fitted (H7 flight controller or a governed ESC), THR_MAX stays at 70 % and the stages that need more throttle are not flown (see the stage table).

FRAME, ORIENTATION AND LOGGING

Board mounted on the wing-bay floor, arrow forward, components up (build guide). Log to the microSD card.

PARAMETER	VALUE	NOTE
AHRS_ORIENTATION	0	FC arrow forward, upright. Change only if the board is mounted otherwise.
LOG_BACKEND_TYPE	1	Logs to the microSD card.
LOG_DISARMED	0	No logging while disarmed.

OUTPUTS

S1 drives the motor with bidirectional DShot (rpm telemetry for the prop-rpm check). S1 and S2 form one timer group: S2 stays unused. S3 drives the single aileron servo (both ailerons through the torque rods). S4 and S5 drive the left and right tail panels as a V-tail mix.

Servo endpoints are calculated for the Emax ES9052MD (about 11.1 us per degree) and the linkages: aileron 2.79:1 (12 mm horn, 4.3 mm arm hole), tail 1.19:1 (8 mm horn, 6.7 mm arm hole). They are starting points: set the final values by measuring the deflections on the bench.

PARAMETER	VALUE	NOTE
SERVO1_FUNCTION	70	S1: throttle (motor ESC).
SERVO2_FUNCTION	0	S2: unused (shares the DShot timer group with S1).
SERVO3_FUNCTION	4	S3: aileron (one servo drives both ailerons).
SERVO4_FUNCTION	79	S4: V-tail left (left tail panel).
SERVO5_FUNCTION	80	S5: V-tail right (right tail panel).
SERVO3_TRIM	1500	Aileron neutral. The 3 deg reflex is set mechanically on each link, not here (one servo moves the ailerons in opposite directions).
SERVO3_MIN	1178	About 10 deg aileron (servo 29 deg at 2.79:1). Verify: about 3.5 mm at the aileron trailing edge.
SERVO3_MAX	1822	As SERVO3_MIN. Linkage limit +-20 deg aileron = +-807 us: never set beyond it.
SERVO4_TRIM	1500	Left panel neutral; trim mechanically first (collar on the shaft flat).

OUTPUTS

PARAMETER	VALUE	NOTE
SERVO4_MIN	1232	+-20 deg panel travel at 1.19:1 (servo 24 deg). Verify by measurement.
SERVO4_MAX	1768	As SERVO4_MIN.
SERVO5_TRIM	1500	Right panel neutral.
SERVO5_MIN	1232	As SERVO4_MIN.
SERVO5_MAX	1768	As SERVO4_MIN.
MIXING_GAIN	0.5	V-tail mix: full pitch alone gives half the panel travel (about +-10 deg), pitch plus yaw can reach the +-20 deg endpoints.
SERVO_RATE	50	Servo update rate (Hz) for the PWM groups; the ES9052MD is a digital servo and also accepts higher rates.

MOTOR, ESC AND RPM TELEMETRY

Bidirectional DShot gives motor rpm in the logs: check on the thrust stand that the unloaded prop stays below 30,000 rpm on a full pack.

SERVO_DSHOT_ESC and the protocol depend on the ESC firmware (AM32 or BLHeli_32): verify on the bench.

PARAMETER	VALUE	NOTE
SERVO_BLH_MASK	1	DShot on output 1.
SERVO_BLH_OTYPE	5	DShot300. Verify the ESC accepts it.
SERVO_BLH_BDMASK	1	Bidirectional DShot (rpm telemetry) on output 1.
SERVO_BLH_POLES	14	Motor magnet poles (12N14P 2207 motors). Verify for your motor.
SERVO_DSHOT_ESC	1	BLHeli_32 / AM32 type ESC. Verify against your firmware version's list.
RPM1_TYPE	5	RPM sensor from ESC telemetry (logged).
RPM1_ESC_MASK	1	Motor on output 1.

THROTTLE

THR_MAX is the prop-rating cap: no governor on the F405 yet: 30,000 / (2550 KV x 16.8 V) = 70 % keeps an unloaded prop on a full pack at its rating. Raise only when an rpm governor is fitted and proven on the thrust stand.

Speed stages are held by a transmitter throttle limit on a switch (LAUNCH / STAGE, table below), not by the flight controller. ArduPilot clips the throttle at THR_MAX in the stabilised modes; check on the bench that MANUAL is clipped too, otherwise fly the stages in FBWA.

PARAMETER	VALUE	NOTE
THR_MAX	70	no governor on the F405 yet: 30,000 / (2550 KV x 16.8 V) = 70 % keeps an unloaded prop on a full pack at its rating. Raise only when an rpm governor is fitted and proven on the thrust stand.
THR_MIN	0	Motor stops at zero throttle in flight (gliding, flutter abort).
THR_SLEWRATE	100	Throttle slew limit (% per second).
TRIM_THROTTLE	35	Cruise throttle guess for the synthetic-airspeed modes. Set from a log of level flight.

NO AIRSPEED SENSOR

No airspeed sensor is fitted. ArduPilot uses a synthetic airspeed from GPS, wind estimate and throttle; FBWA and MANUAL do not need it.

Stall is about 13.0 m/s; AIRSPEED_MIN keeps 25 % margin. AIRSPEED_MAX matches the first speed stage and is raised with the stages.

PARAMETER	VALUE	NOTE
ARSPD_TYPE	0	No airspeed sensor.
ARSPD_USE	0	Do not use an airspeed sensor.
AIRSPEED_MIN	17	m/s; 1.25 x the 13.0 m/s stall.
AIRSPEED_CRUISE	20	m/s; cruise target for CRUISE / FBWB / RTL.
AIRSPEED_MAX	30	m/s; stage 1. Raise with each stage the test plan clears.

ATTITUDE LIMITS, PITCH RATE AND YAW DAMPER

Pitch rate is limited per speed stage so a full pull stays under the 6 g limit (table below); the file holds stage 1 (94 deg/s). Lower it before each faster stage.

The flight-dynamics model finds the Dutch roll about neutral at 13 m/s, so the yaw damper is required. Start low and tune in flight.

PARAMETER	VALUE	NOTE
PTCH_RATE_MAX	94	deg/s, stage 1 (30 m/s). See the stage table.
ROLL_LIMIT_DEG	55	Bank limit in the stabilised modes.
PTCH_LIM_MAX_DEG	25	Nose-up limit.
PTCH_LIM_MIN_DEG	-25	Nose-down limit (no steep powered dives: speed builds fast).
YAW2SRV_DAMP	0.15	Yaw damper gain, starting point. Increase in small steps if the tail wags.
KFF_RDDRMIX	0.5	Rudder (V-tail yaw) mixed with aileron for coordinated turns.

RECEIVER, MODES AND ARMING

RadioMaster RP1 V2 ELRS receiver on UART1 (TX1/RX1) with CRSF. Arm with a switch on channel 5; flight modes on channel 6.

Hand launch in FBWA with the LAUNCH throttle position; no auto-takeoff is needed.

PARAMETER	VALUE	NOTE
SERIAL1_PROTOCOL	23	RCIN (CRSF/ELRS) on UART1.
RC_OPTIONS	8192	Bit 13: 420k baud for ELRS.
RC5_OPTION	153	Channel 5 arms and disarms.
ARMING_RUDDER	0	No stick arming (switch only).
ARMING_CHECK	1	All pre-arm checks.
FLTMODE_CH	6	Flight-mode channel.
FLTMODE1	5	FBWA (launch, stage runs).
FLTMODE2	5	FBWA.
FLTMODE3	0	MANUAL.

RECEIVER, MODES AND ARMING

PARAMETER	VALUE	NOTE
FLTMODE4	0	MANUAL.
FLTMODE5	11	RTL.
FLTMODE6	11	RTL.

GPS AND COMPASS

TBS M10Q on UART3 (TX3/RX3, the GPS pads) with its QMC5883 compass on I2C (SDA/SCL). The GPS sits behind the battery: its compass is 3-6 cm from the battery leads.

Either calibrate and run compass-motor interference compensation (compassmot) on the thrust stand, or disable the compass and fly on GPS heading (COMPASS_USE 0).

PARAMETER	VALUE	NOTE
SERIAL3_PROTOCOL	5	GPS on UART3.
GPS_TYPE	1	Auto (u-blox M10). Named GPS1_TYPE in ArduPlane 4.6+.
COMPASS_ENABLE	1	Use the M10Q's compass (set 0 to fly without).
COMPASS_AUTODEC	1	Automatic declination.
COMPASS_AUTO_ROT	2	Detect the compass orientation during calibration. Verify the result against the GPS's arrow.

BATTERY MONITOR AND FAILSAFES

WING MINI PDB sensors: voltage divider 1K:10K, current 50 A/V. Calibrate both against a meter.

Radio failsafe: RTL after 5 s without signal. Low battery: RTL.

PARAMETER	VALUE	NOTE
BATT_MONITOR	4	Analog voltage and current.
BATT_VOLT_PIN	10	PDB voltage.
BATT_CURR_PIN	11	PDB current.
BATT_VOLT_MULT	11.05	Calibrate against a meter.
BATT_AMP_PERVLT	50	Calibrate against a meter or a known charge.
BATT_CAPACITY	650	mAh (4S 650).
BATT_LOW_VOLT	13.6	3.4 V per cell under load.
BATT_LOW_MAH	160	About 25 % left.
BATT_FS_LOW_ACT	1	RTL on low battery.

BATTERY MONITOR AND FAILSAFES

PARAMETER	VALUE	NOTE
THR_FAILSAFE	1	Radio failsafe on.
FS_SHORT_ACTN	0	Short failsafe: circle (default).
FS_LONG_ACTN	1	Long failsafe: RTL.
FS_LONG_TIMEOUT	5	Seconds.

OSD AND WIRELESS TELEMETRY

Analog OSD (board OSD chip) with ground speed, throttle, current, used capacity and voltage. ArduPilot has no ground-speed alarm: watch the ground-speed number against the stage limit and judge speed from two-way passes.

DJI O4 noses: switch to MSP DisplayPort on UART5 (SERIAL5_PROTOCOL 42, OSD_TYPE 5).

The wireless USB extender is on UART6 (MAVLink2, 115200) for setup on the ground.

PARAMETER	VALUE	NOTE
OSD_TYPE	1	Analog OSD. Use 5 (MSP DisplayPort) with the DJI O4 noses.
OSD1_ENABLE	1	Screen 1.
OSD1_GSPEED_EN	1	GPS ground speed.
OSD1_THROTTLE_EN	1	Throttle %.
OSD1_CURRENT_EN	1	Current.
OSD1_BATUSED_EN	1	Used capacity (mAh).
OSD1_BAT_VOLT_EN	1	Battery voltage.
OSD_W_BATVOLT	14.0	Battery voltage warning (V).
SERIAL6_PROTOCOL	2	MAVLink2 on UART6 (wireless board).

OSD AND WIRELESS TELEMETRY

PARAMETER	VALUE	NOTE
SERIAL6_BAUD	115	115200 baud.

BENCH CHECKS BEFORE FLIGHT

- Props off. Power from a full pack; check the battery voltage and current readings against a meter.
- Control directions in MANUAL: right stick -> right aileron trailing edge up, left down. Pull -> both tail-panel trailing edges up. Right rudder -> left panel trailing edge down, right panel up (inverted V, panels below the fuselage). Fix with SERVOx_REVERSED.
- Measure deflections with a protractor: aileron about +-10 deg, tail panels +-20 deg at the endpoints; adjust SERVOx_MIN/MAX. Nothing binds or buzzes at the ends.
- FBWA: roll and pitch the airframe by hand; the surfaces must oppose the motion.
- Thrust stand: THR_MAX 70 %; logged rpm with the prop unloaded on a full pack below 30,000; ESC current within its rating.
- Transmitter: LAUNCH / STAGE switch set to the stage limit; confirm in the servo-output screen that full stick on STAGE gives the stage %.
- Radio failsafe: switch the transmitter off with the motor at low throttle (props off): motor stops or RTL behaviour as set.
- Compass: calibrate, then compassmot on the thrust stand (or COMPASS_USE 0).
- GPS: 3D fix and the OSD shows ground speed.